

# Report Notes

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Content for the CT050-3-2-WAPP group report. Paste each section into the university template; diagrams are Mermaid (render them on GitHub or export them as images) and the screenshots used below are the ones the presentation site serves from `FirestoreLanding/assets/screens/`.

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## Title page

| Title             | LearnHub – A Web-Based Learning Management System                                                  |
|-------------------|----------------------------------------------------------------------------------------------------|
| Module            | CT050-3-2-WAPP Web Applications                                                                    |
| Assessment        | Group assignment                                                                                   |
| Institution       | Asia Pacific University of Technology & Innovation (APU)                                           |
| Team members      | Khaytboy Khayrullaev (TP072305) — carried out individually                                         |
| Lecturer          | <i>Lecturer's name</i>                                                                             |
| Intake            | <i>Intake code</i>                                                                                 |
| Submission date   | <i>Date</i>                                                                                        |
| Repository        | <a href="https://github.com/KhayitOff/LearnHub">https://github.com/KhayitOff/LearnHub</a>          |
| Presentation site | <a href="https://learnhub-wapp.web.app">https://learnhub-wapp.web.app</a> (Firebase Hosting, live) |

## Abstract

LearnHub is a web-based learning management system for computing students, built with ASP.NET Core MVC on .NET 10, C# and Razor views, Bootstrap 5.3, Entity Framework Core 10 and ASP.NET Core Identity. Guests explore a catalogue of courses and free preview lessons; registered students enrol, study articles, videos, PDF documents and images, mark lessons complete, take quizzes with explained answers and follow their progress on a data-driven dashboard. Administrators manage courses, categories, learning resources, quizzes, enrolments, users and contact messages in a protected admin area. One SQLite database with a single code-first migration set serves every environment, and on the hosting platform its file lives on a mounted volume so it survives each redeploy. Security follows the OWASP Top 10:2025: role-based authorisation, antiforgery tokens, encoded output with a strict Content Security Policy, ownership checks, validated uploads, rate limiting and no committed credentials. Quality is assured by 211 automated .NET tests, 42 browser and accessibility checks on three screen sizes, and continuous integration that also builds the Docker image and runs it the way the host will. The static presentation site is live on Firebase Hosting and links to the application, because Firebase cannot execute ASP.NET Core.

# 1. Introduction

## 1.1 Background

Students learning programming, web development, databases, security, networking and cloud computing usually rely on scattered resources: slides, videos, articles and cheat sheets. The material rarely has a clear order, gives no feedback on understanding and offers no view of progress. Lecturers who want to share structured material need a tool that is simple to maintain and safe for students to use.

## 1.2 Problem statement

1. Learners do not know what to study next or how far they are from finishing a topic.
2. Reading and watching do not show whether a topic was understood.
3. Content managers lack one secure place to publish, update and retire courses and to see learner activity.

## 1.3 Proposed solution

LearnHub organises learning into courses made of ordered lessons and quizzes. Guests can explore freely; students enrol, study and receive immediate, explained quiz feedback; progress is calculated from real activity and shown on a dashboard. Administrators use a separate, protected area to manage everything. The whole interface follows one metaphor: LearnHub is an **interchange** where each category is a coloured line, each course is a route and each lesson is a stop.

## 1.4 Report structure

Section 2 covers planning. Section 3 presents the design (architecture, use cases, flowcharts, wireframes, navigation and database). Section 4 explains the implementation. Sections 5 and 6 describe testing and deployment, section 7 walks through screenshots, and sections 8 to 10 evaluate the work, propose enhancements and conclude.

# 2. Project planning

## 2.1 Mission statement

LearnHub gives every computing student a clear route through each subject, with honest feedback at every stop, and gives administrators one simple place to keep that route up to date.

## 2.2 Objectives

| #  | Objective                                                          | Achieved | Evidence                                                       |
|----|--------------------------------------------------------------------|----------|----------------------------------------------------------------|
| O1 | Public catalogue with search, filters, details and preview lessons | Yes      | Public site tests; screenshots 7.1–7.3                         |
| O2 | Continuous student journey: register, enrol, study, quiz, result   | Yes      | <code>StudentJourneyTests</code> ; Playwright student scenario |

| #  | Objective                                                           | Achieved                                                                                  | Evidence                                                                                 |
|----|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| O3 | Data-driven student dashboard                                       | Yes                                                                                       | Dashboard figures, progress, results, activity, recommendations (7.5)                    |
| O4 | Full admin CRUD for content, enrolments and users                   | Yes                                                                                       | <code>AdminManagementTests</code> ; admin browser tests                                  |
| O5 | Protection against OWASP Top 10 risks                               | Yes                                                                                       | Security review; attack-style automated tests                                            |
| O6 | Responsive and accessible on phone, tablet and desktop              | Yes                                                                                       | 42 browser checks at three sizes; no WCAG 2.2 A/AA violations from axe-core              |
| O7 | Automated testing and container deployment with a presentation site | Partly – Firebase Hosting is live; the Railway service is configured but not yet deployed | Three CI jobs; <a href="https://learnhub-wapp.web.app">https://learnhub-wapp.web.app</a> |

## 2.3 Audience modelling

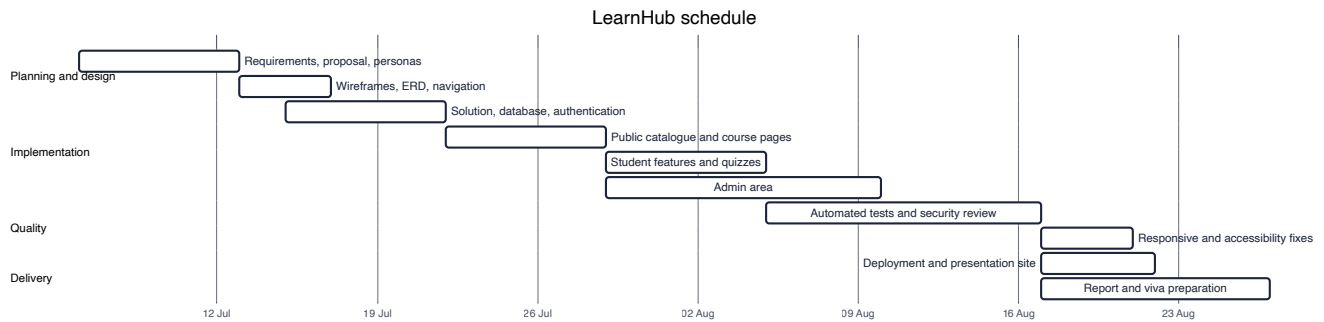
| Persona                               | Profile                                                 | Needs                                                        | How LearnHub responds                                                                                         |
|---------------------------------------|---------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Nurul, first-year student             | 19, Diploma in IT, studies on a mid-range Android phone | A clear order of topics; checking understanding before tests | Ordered course routes, a Resume button, quizzes with explanations, mobile-first layout                        |
| Daniel, career switcher               | 31, full-time job, studies in the evenings on a laptop  | Practical skills and visible progress in limited time        | Difficulty and duration on every course, progress bars, "Continue learning", PDF cheat sheets                 |
| Ms. Priya, lecturer and administrator | 42, manages networking and cloud modules on a desktop   | Quick publishing, safe editing, overview of learner activity | Admin dashboard, publish/unpublish, impact-aware delete confirmations, validated uploads, quiz result reports |

## 2.4 Scope

In scope: public catalogue and pages, registration and login, student dashboard, enrolment, lessons of five types, completion tracking, quizzes and results, profile management, a complete admin area, security controls, responsive and accessible design, automated tests, CI/CD, a Docker image that runs on Railway and a Firebase Hosting presentation site.

Out of scope for version 1.0: payments, certificates, chat, forums, live classes, an AI tutor, an instructor self-service role, email delivery (confirmation and password reset), native apps and multiple languages.

## 2.5 Schedule



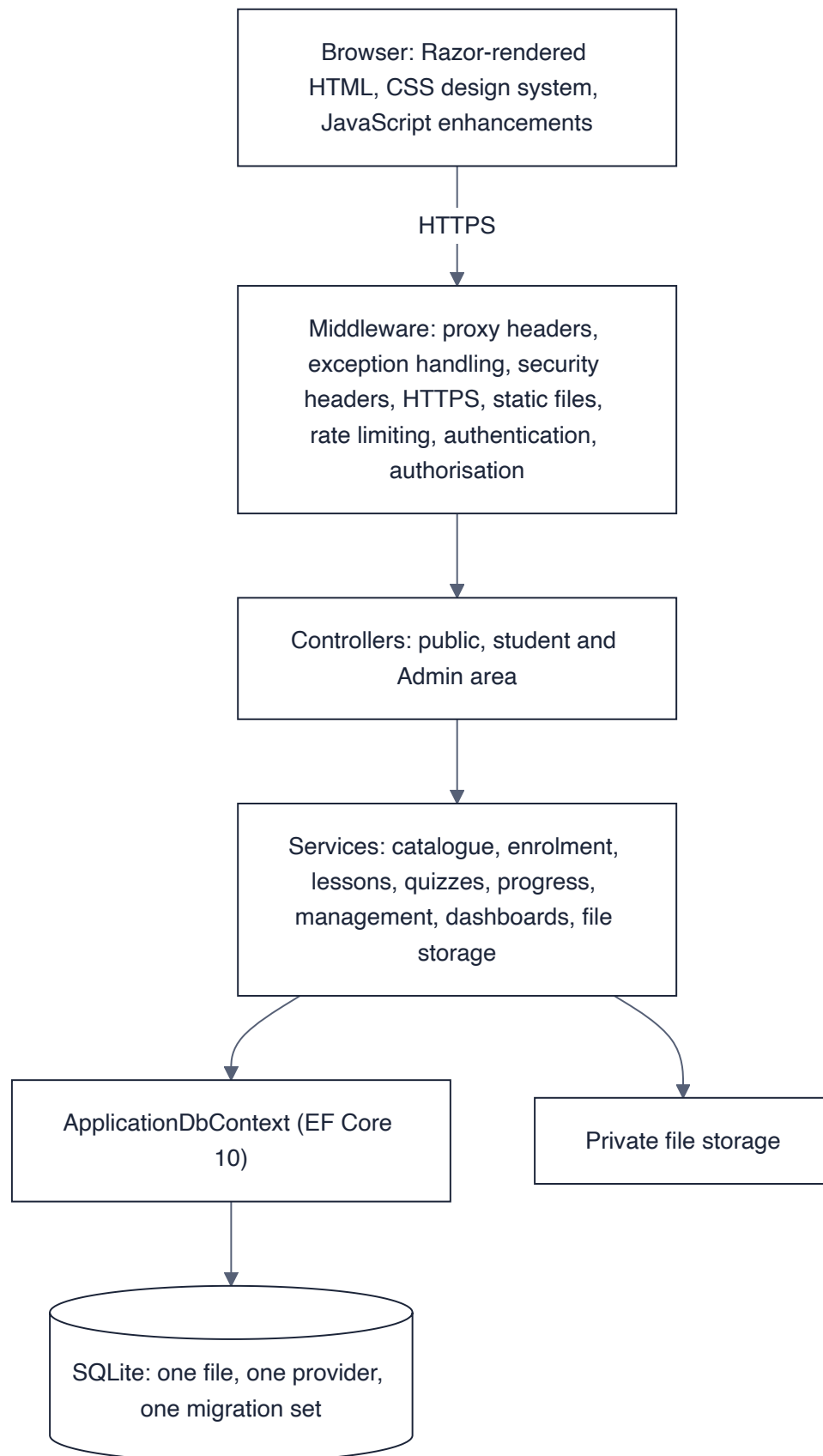
## 2.6 Team roles

This submission was completed by one member working alone, so a single person covered every role: architecture, database design, authentication, the public site, the student area, the administration area, the security review, the automated tests, the documentation and the deployment. The areas covered and the way the work was organised are set out in [TEAM.md](#); the full proposal is in [PROPOSAL.md](#).

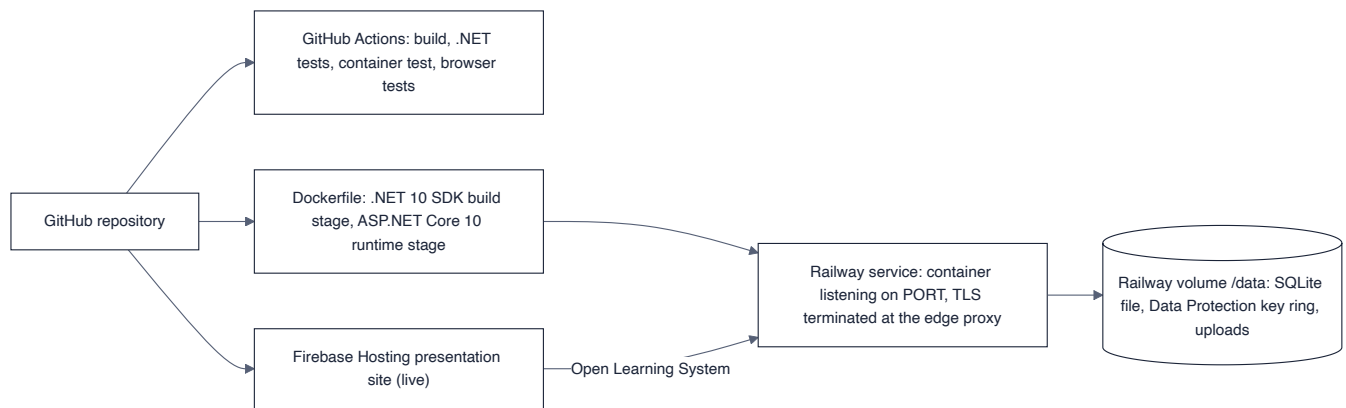
## 3. System design

### 3.1 Architecture

LearnHub is a single ASP.NET Core MVC application with clear layers. Controllers handle HTTP concerns, services contain business rules and queries, and Entity Framework Core accesses the database. Views receive view models only, and forms bind to view models, never to entities.



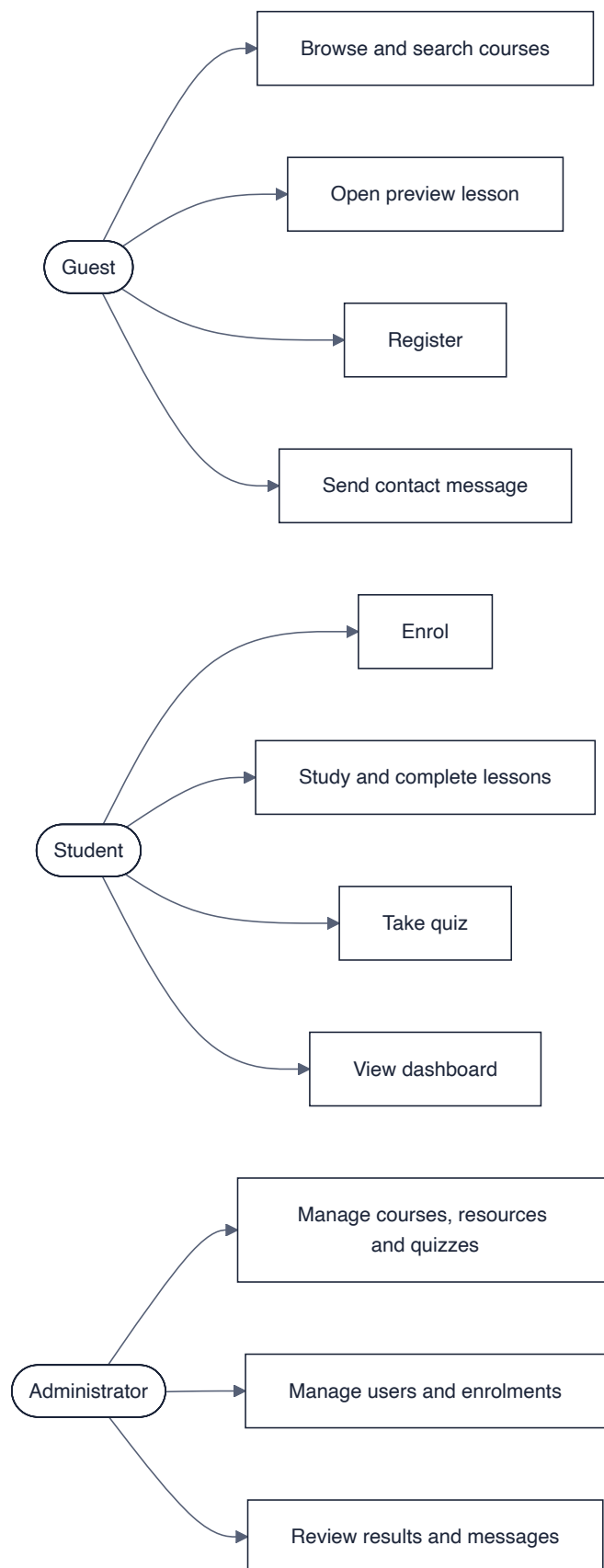
Deployment view:



The detailed design, including every route, is in [ARCHITECTURE.md](#).

### 3.2 Use cases

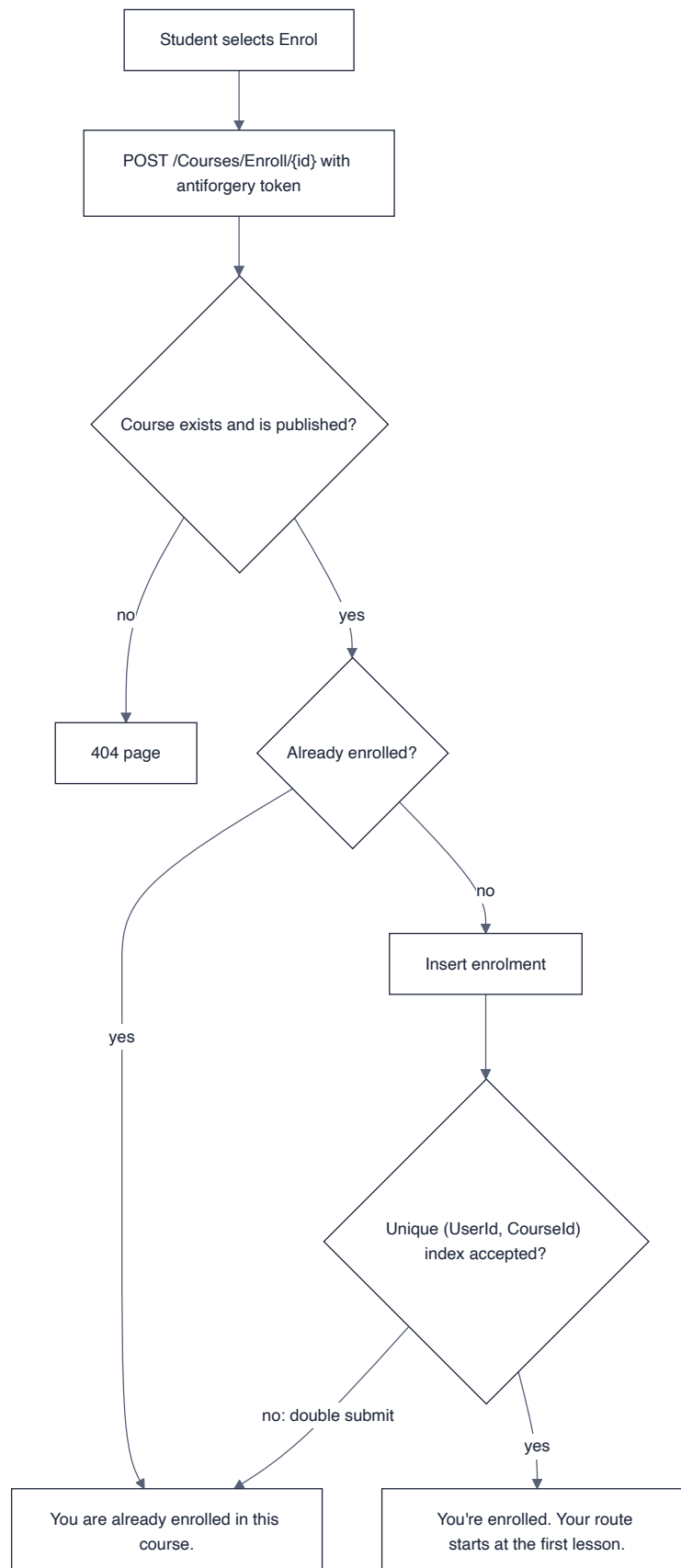
| ID       | Use case                                                                                                                | Actor                         |
|----------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| UC-01–02 | Register, log in and log out                                                                                            | Guest, student, administrator |
| UC-03–05 | Browse, search and filter courses; view details; open a preview lesson                                                  | Guest                         |
| UC-06    | Send a contact message                                                                                                  | Guest                         |
| UC-07–09 | Enrol, leave, study lessons and mark them complete                                                                      | Student                       |
| UC-10–13 | Take quizzes, review results, use the dashboard, manage profile                                                         | Student                       |
| UC-14–22 | Admin dashboard; manage courses, categories, resources, quizzes and questions, enrolments, users, attempts and messages | Administrator                 |



Every use case has preconditions, main and alternative flows and its implementing controller in [USE\\_CASES.md](#).

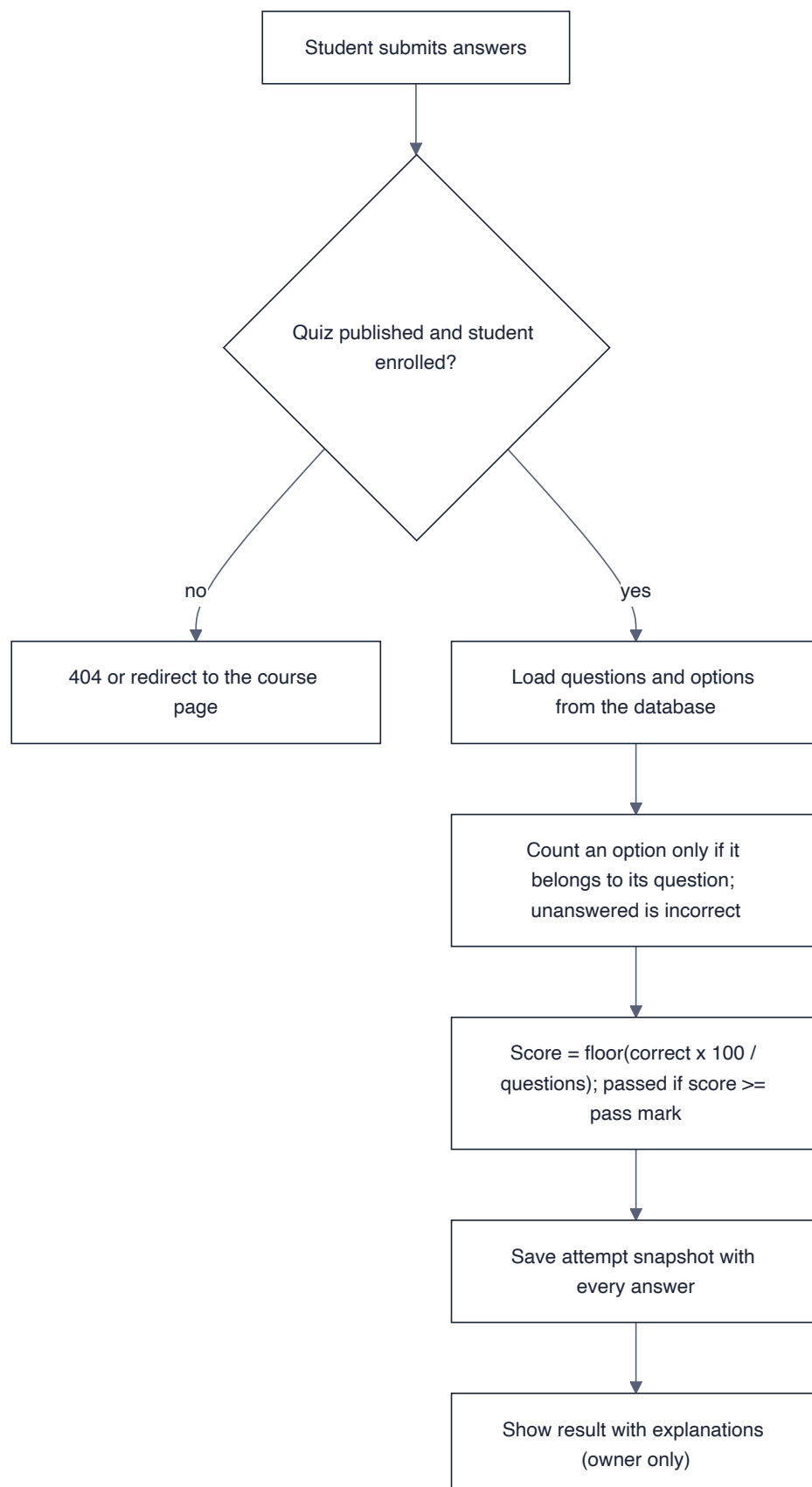
### 3.3 Flowcharts

Enrolment (from `CoursesController.Enroll` and `EnrollmentService`):



Quiz submission and grading (from `QuizzesController`, `QuizService` and `QuizGrader`):



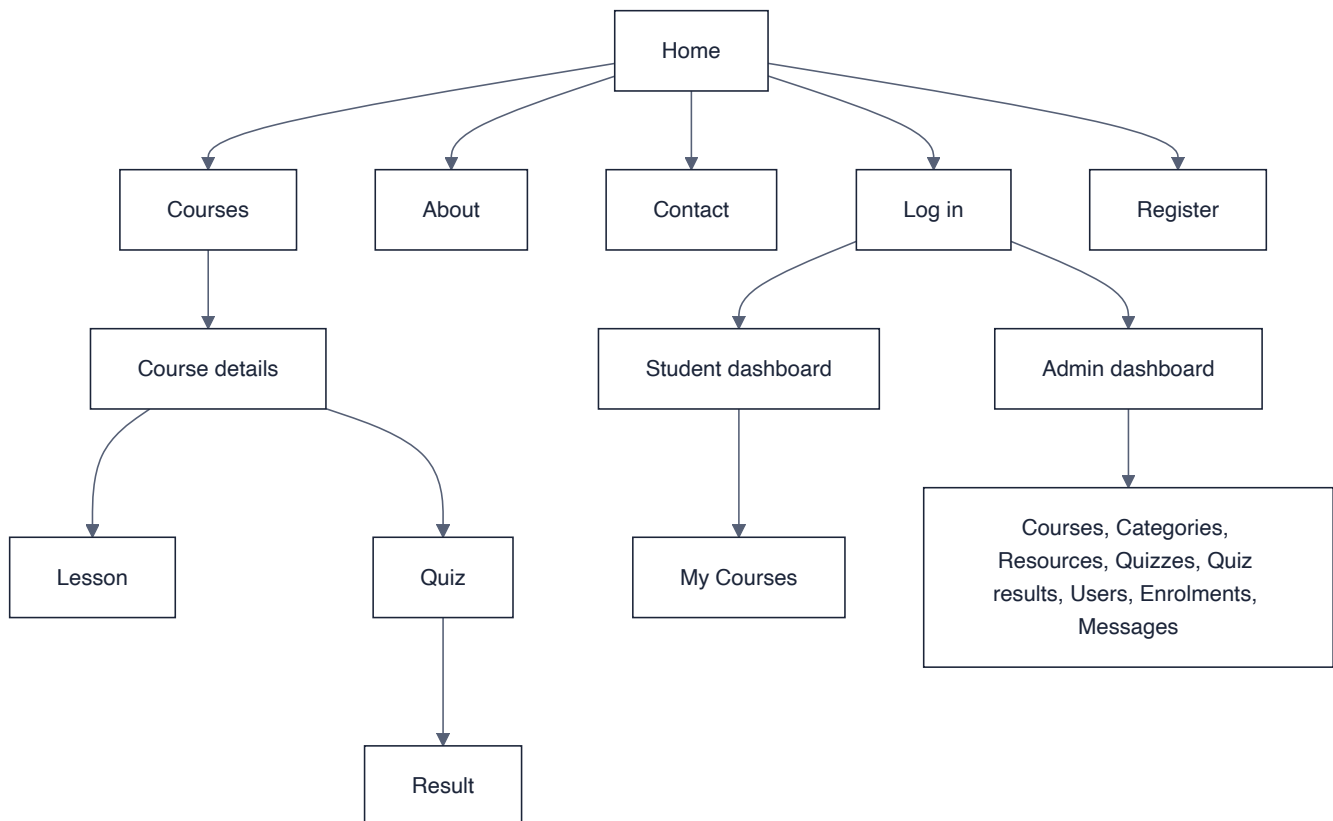


Eleven flowcharts, including the request pipeline, login, lesson access, course deletion and upload validation, are in [FLOWCHARTS.md](#).

### 3.4 Wireframes

Low-fidelity wireframes were produced for the home page, catalogue, course details, lesson player, quiz and result, student dashboard, authentication forms, admin dashboard, admin list pages and admin forms, each in desktop and phone variants. They fixed the key layout decisions early: a two-column hero with the category map, a sidebar of coloured category filters in the catalogue, a sticky enrolment panel beside the course route, a sticky lesson outline, and an admin shell with a dark sidebar that becomes an off-canvas menu on small screens. See [WIREFRAMES.md](#).

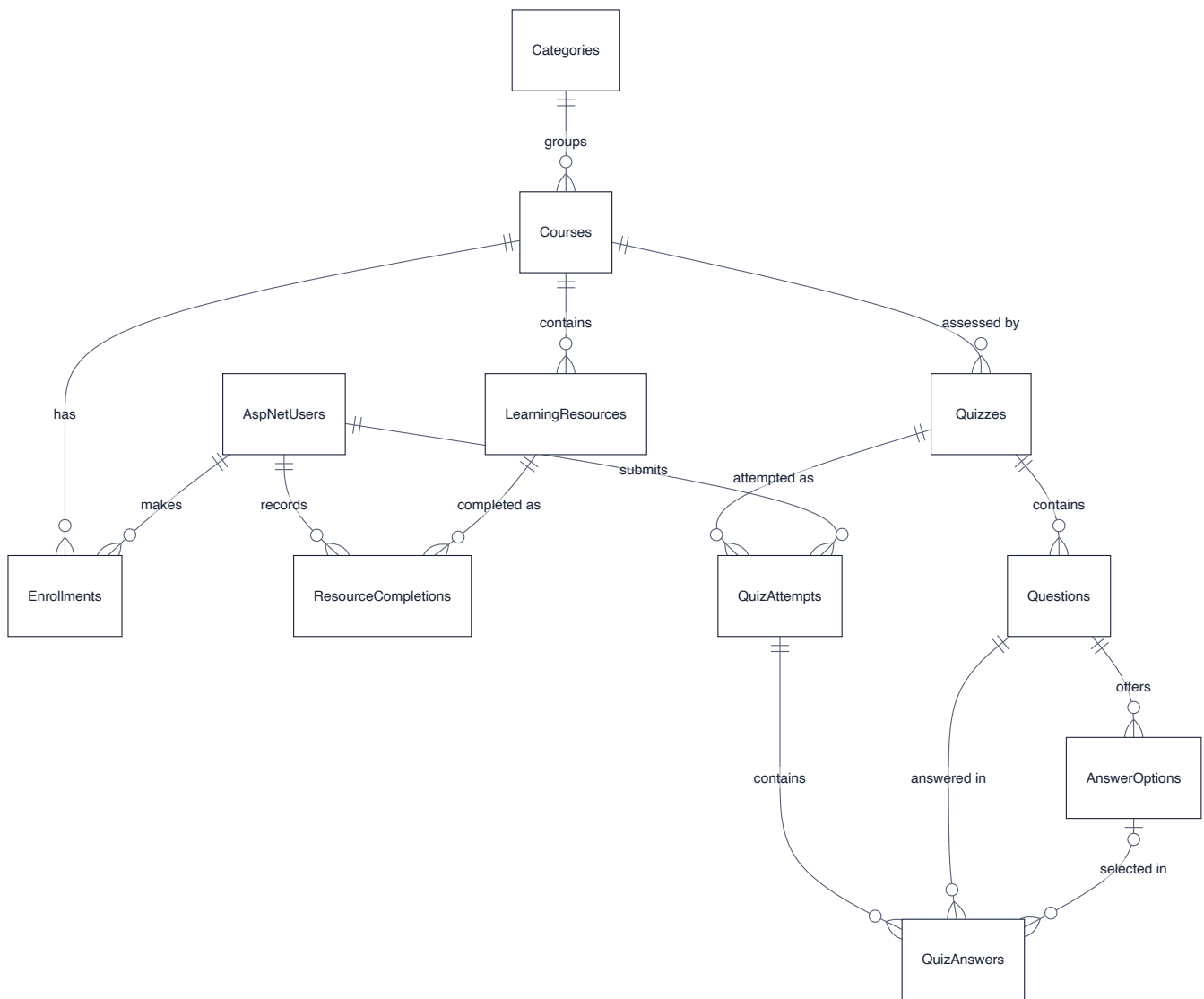
### 3.5 Navigation structure



Guests see Home, Courses, About, Contact, Log in and Register; students also see My Courses, Dashboard and a user menu; administrators see an Admin link and use the admin sidebar. Breadcrumbs appear on nested pages. The complete site map, route table and redirect behaviour are in [NAVIGATION.md](#).

### 3.6 Database design

The database has eleven application tables — `Categories`, `Courses`, `LearningResources`, `Enrollments`, `ResourceCompletions`, `Quizzes`, `Questions`, `AnswerOptions`, `QuizAttempts`, `QuizAnswers` and `ContactMessages` — plus the seven ASP.NET Core Identity tables. One code-first migration creates all eighteen tables, and SQLite is the only provider, so there is one context and one migration set to keep in step rather than one schema per dialect.



### Key design decisions:

- **Integrity:** primary and foreign keys, required columns with lengths from one `FieldLengths` class, unique indexes (category name; one enrolment per student and course; one completion per student and resource; one answer per question per attempt) and check constraints (positive course duration, completion percentage, pass mark and score inside their ranges, question points between one and one hundred, correct count never above the question count).
- **Delete behaviour:** deleting a category is restricted while courses exist; a course cascades to its resources, quizzes, attempts and enrolments; and quiz answers restrict deletion of the questions and options they refer to, so editing a quiz can never silently erase a student's answer history — the services delete the dependent answers first, inside one transaction.
- **Truthful history:** an attempt stores a snapshot of `CorrectCount`, `QuestionCount`, `Score`, `MaxScore`, `ScorePercent` and `Passed`; progress is calculated rather than stored; deactivation uses Identity lockout instead of a separate flag.
- **Location:** the connection string is read from `DATABASE_CONNECTION_STRING`, falling back to `ConnectionStrings:DefaultConnection`; a relative path is resolved against the content root and its folder is created automatically, and on the hosting platform the path points at the persistent volume (`Data Source=/data/learnhub.db`).

- **Time:** all dates are stored in UTC with automatic `CreatedAt` and `UpdatedAt`.

The full diagram with every column, index and constraint is in [ERD.md](#), and the reasoning behind each decision is in [DATABASE.md](#).

## 4. Implementation

### 4.1 HTML5 and Razor

Pages use semantic HTML5 landmarks ( `header`, `nav`, `main`, `section`, `aside`, `footer` ), headings in order, and labelled forms. Razor layouts ( `Views/Shared/_Layout.cshtml`, `Areas/Admin/Views/Shared/_AdminLayout.cshtml` ) provide the page frame; partials such as `_CourseCard`, `_Pagination` and `_QuizReview` avoid repetition. Built-in tag helpers generate URLs, validation attributes and antiforgery tokens, and two custom tag helpers render navigation state and course covers.

```
<div class="mb-3">
  <label asp-for="Password" class="form-label"></label>
  <input asp-for="Password" class="form-control" autocomplete="new-password" data-password-meter="password-meter" aria-describedby="password-help" />
  <div class="password-meter" id="password-meter" data-score="0" aria-hidden="true"><span>
</span><span></span><span></span><span></span></div>
  <div class="form-text" aria-live="polite"></div>
  <div id="password-help" class="form-text field-hint">@PasswordRules.Description</div>
  <span asp-validation-for="Password"></span>
</div>
```

Figure: password field in `Views/Account/Register.cshtml` – the label, input type, maximum length and validation attributes all come from the view model through tag helpers.

### 4.2 CSS3 and the design system

`wwwroot/css/site.css` defines the design system as CSS custom properties and builds every component from them; the admin shell in `admin.css` reuses the same tokens.

```
--lh-ink: #1b2438;
--lh-paper: #f5f7fa;
--lh-surface: #ffffff;
--lh-rail: #dde3eb;
--lh-muted: #566076;
--lh-signal: #ffc53d;
--lh-line-0: #4b5a73;
--lh-line-1: #c9344a;
--lh-line-2: #157f8e;
--lh-font: "Overpass", "Segoe UI", system-ui, -apple-system, Roboto, "Helvetica Neue", Arial, sans-serif;
```

Figure: an excerpt of the design tokens in `wwwroot/css/site.css`.

Each category receives one of eight line colours from its id, used on course covers, category filters, course routes and the home page map. Layouts are mobile-first with CSS Grid and Flexbox, and key responsive decisions are documented in the wireframes. Accessibility is built in: a visible yellow focus halo on every interactive element, skip links, contrast checked against WCAG AA, reduced-motion support, and focusable scroll regions for wide code and tables.

## 4.3 JavaScript

JavaScript is used only for progressive enhancement, so every feature still works when it is disabled.

Examples in `wwwroot/js/site.js`: local date formatting, character counters, a password strength meter, image previews before upload, type-specific fields on the resource form, the answer option editor, confirmation prompts and the quiz helper below.

```
const answeredCount = () => questions.filter((q) => q.querySelector("input:checked")).length;
const update = () => {
  if (counter) {
    counter.textContent = `${answeredCount()} of ${questions.length} answered`;
  }
};

form.addEventListener("change", update);
form.addEventListener("submit", (event) => {
  const unanswered = questions.length - answeredCount();
  if (unanswered > 0) {
    const noun = unanswered === 1 ? "question is" : "questions are";
    if (!window.confirm(`${unanswered} ${noun} unanswered and will be marked incorrect.
Submit anyway?`)) {
      event.preventDefault();
    }
  }
});
```

Figure: the quiz answered counter and unanswered warning in `wwwroot/js/site.js`.

## 4.4 Validation

Validation is defined once on view models and enforced twice: in the browser through jQuery Validation Unobtrusive and on the server through model binding and `ModelState`.

```
[Required(ErrorMessage = "Please choose a password.")]
[StringLength(PasswordRules.MaximumLength, MinimumLength = PasswordRules.MinimumLength,
ErrorMessage = "Password must be between {2} and {1} characters.")]
[RegularExpression(PasswordRules.Pattern, ErrorMessage = PasswordRules.Description)]
[DataType(DataType.Password)]
public string Password { get; set; } = string.Empty;

[Required(ErrorMessage = "Please confirm your password.")]
[Compare(nameof(Password), ErrorMessage = "The passwords do not match.")]
[DataType(DataType.Password)]
```

Figure: password rules in `ViewModels/Account/AccountViewModels.cs`.

Custom attributes extend this: `MustBeTrueAttribute` requires the terms checkbox, and `MaxFileSizeAttribute` and `AllowedExtensionsAttribute` check uploads; all three also emit client-side rules. Rules that need the database are checked in services: unique category names, two to six answer options with exactly one correct, a question before publishing a quiz, enrolment only in published courses and protection of the last administrator. Upload validation on the server goes further than the browser can: it checks the file signature and the real number of bytes written.

## 4.5 Database and Entity Framework Core

`ApplicationDbContext` holds the whole model, and one provider is registered for every environment: SQLite, in `Data/DatabaseServiceCollectionExtensions.cs`. There is a single migrations folder, and `Data/DesignTimeDbContextFactory.cs` is the only design-time factory `dotnet ef` needs. Relationships and constraints use the Fluent API:

```
public void Configure(EntityTypeBuilder<Enrollment> builder)
{
    // A student can enrol in a course only once.
    builder.HasIndex(e => new { e.UserId, e.CourseId }).IsUnique();

    builder.HasOne(e => e.User)
        .WithMany(u => u.Enrollments)
        .HasForeignKey(e => e.UserId)
        .OnDelete(DeleteBehavior.Cascade);

    builder.HasOne(e => e.Course)
        .WithMany(c => c.Enrollments)
        .HasForeignKey(e => e.CourseId)
        .OnDelete(DeleteBehavior.Cascade);
}
```

Figure: enrolment configuration in `Data/Configurations/UserAndCatalogConfigurations.cs`.

Queries use LINQ with projections to view models and `AsNoTracking` for reading. Multi-step changes run inside the provider's execution strategy, so the unit stays correct if a retrying strategy is ever configured:

```
public static Task InTransactionAsync(this ApplicationDbContext db, Func<Task> operation,
CancellationToken cancellationToken = default)
{
    var strategy = db.Database.CreateExecutionStrategy();
    return strategy.ExecuteAsync(async () =>
    {
        await using var transaction = await
db.Database.BeginTransactionAsync(cancellationToken);
        await operation();
        await transaction.CommitAsync(cancellationToken);
    });
}
```

Figure: `Data/TransactionExtensions.cs`.

At start-up `DatabaseInitializer` applies pending migrations, creates the `Admin` and `Student` roles, creates the administrator from configuration and, for an empty database, seeds the demonstration catalogue: 8 categories, 12 courses (11 published and 1 draft), 64 learning resources (including original PDF cheat sheets and diagrams), 11 quizzes, 54 questions and 216 answer options, together with 7 learners whose enrolments, completions and attempts make the dashboards and progress pages realistic. All four data operations are demonstrated: insert (enrolments, attempts, admin create forms), display (catalogue, dashboards), update (edit forms, completion toggles) and delete (admin delete pages, leaving a course).

## 4.6 Authentication and authorisation

ASP.NET Core Identity provides password hashing, logout (five failures, fifteen minutes), roles and cookie authentication. The cookie is `HttpOnly`, `SameSite=Lax` and `Secure` in production, and the security stamp is re-validated every five minutes so role changes and deactivations take effect quickly. A claims factory adds the user's full name for the interface.

```
[Area("Admin")]
[Authorize(Roles = AppRoles.Admin)]
public abstract class AdminControllerBase : Controller
{
    protected string CurrentUserId => User.GetRequiredUserId();

    protected void Success(string message) => TempData.SetStatus(message);

    protected void Failure(string message) => TempData.SetStatus(message, StatusKind.Danger);
}
```

Figure: every admin controller inherits this base class  
(`Areas/Admin/Controllers/AdminControllerBase.cs`).

Student controllers carry `[Authorize(Roles = AppRoles.Student)]`. Login redirects only to local return URLs. Administrators cannot change, deactivate or delete their own account, and the last active administrator is protected. No password is stored in the repository: in Development a strong random password is generated into the git-ignored `App_Data/demo-credentials.json`, locally it can be overridden with user secrets, and in production it must come from the `Seed__AdminPassword` variable.

## 4.7 Security

| Threat                         | Control in LearnHub                                                                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Broken access control and IDOR | Role attributes, admin base class, ownership and enrolment checks in services, files streamed after access checks, 404 for other users' results                  |
| Cross-site request forgery     | Global antiforgery validation on every unsafe request; state changes only through POST forms                                                                     |
| Cross-site scripting           | Razor encoding, encode-first lesson renderer, strict Content Security Policy without inline scripts                                                              |
| SQL injection                  | EF Core parameterised queries; escaped LIKE patterns                                                                                                             |
| Overposting                    | View models only                                                                                                                                                 |
| Malicious uploads              | Extension allow-list, size limits, file signature checks, random names, storage outside <code>wwwroot</code>                                                     |
| Brute force and spam           | Lockout, generic login errors, rate limiting, contact form honeypot                                                                                              |
| Misconfiguration               | HTTPS and HSTS, security headers, friendly error pages, secure cookies                                                                                           |
| Secrets exposure               | User secrets locally; placeholder-only <code>.env.example</code> with a git-ignored <code>.env</code> ; hosting variables in production; no demo password in Git |

The review against the OWASP Top 10:2025 with test evidence is in [SECURITY.md](#).

## 4.8 Multimedia

- **Images:** ten original SVG course covers built from each category's line colour and an icon, PNG teaching diagrams (for example the MVC request flow, CSS box model and OSI model), and administrator-uploaded course covers.
- **Video:** lessons embed YouTube or Vimeo videos through a strict URL parser and privacy-enhanced players, allowed by the CSP `frame-src` directive.
- **Documents:** original PDF cheat sheets (C#, HTTP status codes, SQL joins, OWASP Top 10:2025, subnetting) open inline for enrolled students through a protected download action.
- **Interactive graphics:** the home page category map is an accessible SVG whose lines are links.

## 4.9 Error handling and logging

Production uses the exception handler and status code pages, so users see branded 404, 403 and 500 pages without technical details, while developers get the detailed exception page locally. Services turn expected problems (not found, validation, unique-index races) into `OperationResult` values and friendly messages. Structured logging records important events such as registrations, sign-ins and blocked sign-ins,



administrator creation, enrolments, quiz submissions, deletions and rejected uploads. A `/health` endpoint checks the database connection for monitoring and smoke tests.

## 5. Testing

LearnHub combines several test levels, all run by GitHub Actions on every push:

| Level                                                                         | Scope                                                                                                                                                                 | Result on 14 September 2026                                                    |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Unit, service and integration (xUnit v3, <code>WebApplicationFactory</code> ) | Grading, validation, content safety, uploads, catalogue and quiz services, public site, authentication, authorisation, student journey, admin CRUD, link crawl        | 211 of 211 passed on SQLite                                                    |
| Container (Docker image, production settings)                                 | The image the host will run, started with <code>PORT=8080</code> and a volume mounted at <code>/data</code> , then smoke tested, checked for clean logs and restarted | Passed; migrations were applied once and were not re-applied after the restart |
| Browser (Playwright)                                                          | 14 scenarios — public pages, the student journey, the admin journey and accessibility scans — on desktop, tablet and phone                                            | 42 of 42 passed (14 scenarios × 3 viewports)                                   |
| Accessibility (axe-core)                                                      | WCAG 2.2 A and AA rules on 26 page views per screen size, inside the browser suite                                                                                    | 9 of 9 passed, no violations                                                   |

Representative test cases:

| Test case                                           | Expected                                                                   | Result |
|-----------------------------------------------------|----------------------------------------------------------------------------|--------|
| Register with a weak password                       | Field-level error with the password rule                                   | Pass   |
| Five wrong passwords                                | Account locked for 15 minutes                                              | Pass   |
| Login with an external return URL                   | Redirect to the user's own dashboard                                       | Pass   |
| Guest opens a locked lesson                         | Redirect to login                                                          | Pass   |
| Enrol twice (including simultaneous requests)       | "You are already enrolled in this course."                                 | Pass   |
| Submit a quiz option from another question          | Counted as incorrect                                                       | Pass   |
| Student opens another student's result              | 404                                                                        | Pass   |
| Student posts to an admin action with a valid token | Refused                                                                    | Pass   |
| Upload HTML disguised as a PDF                      | Rejected; nothing stored                                                   | Pass   |
| Delete a category that still has courses            | Refused with an explanation                                                | Pass   |
| Delete a course with quiz attempts                  | Everything removed in order, with the dependent quiz answers deleted first | Pass   |

| Test case                                    | Expected                         | Result |
|----------------------------------------------|----------------------------------|--------|
| Crawl every page as guest, student and admin | No broken links or server errors | Pass   |

Testing found and fixed defects that no single level would have caught, including a query that SQLite could not translate, a relationship error when deleting courses, form pages that failed behind a TLS-terminating proxy until forwarded headers were configured explicitly, a migration that was never regenerated after a model change and broke 126 tests until it was recreated, pages that scrolled sideways on phones and tablets, an invisible admin sidebar, overlapping map labels, two accessibility problems and 91 build warnings in the test code. The strategy, full inventory, defect log and manual checklist are in [TESTING.md](#).

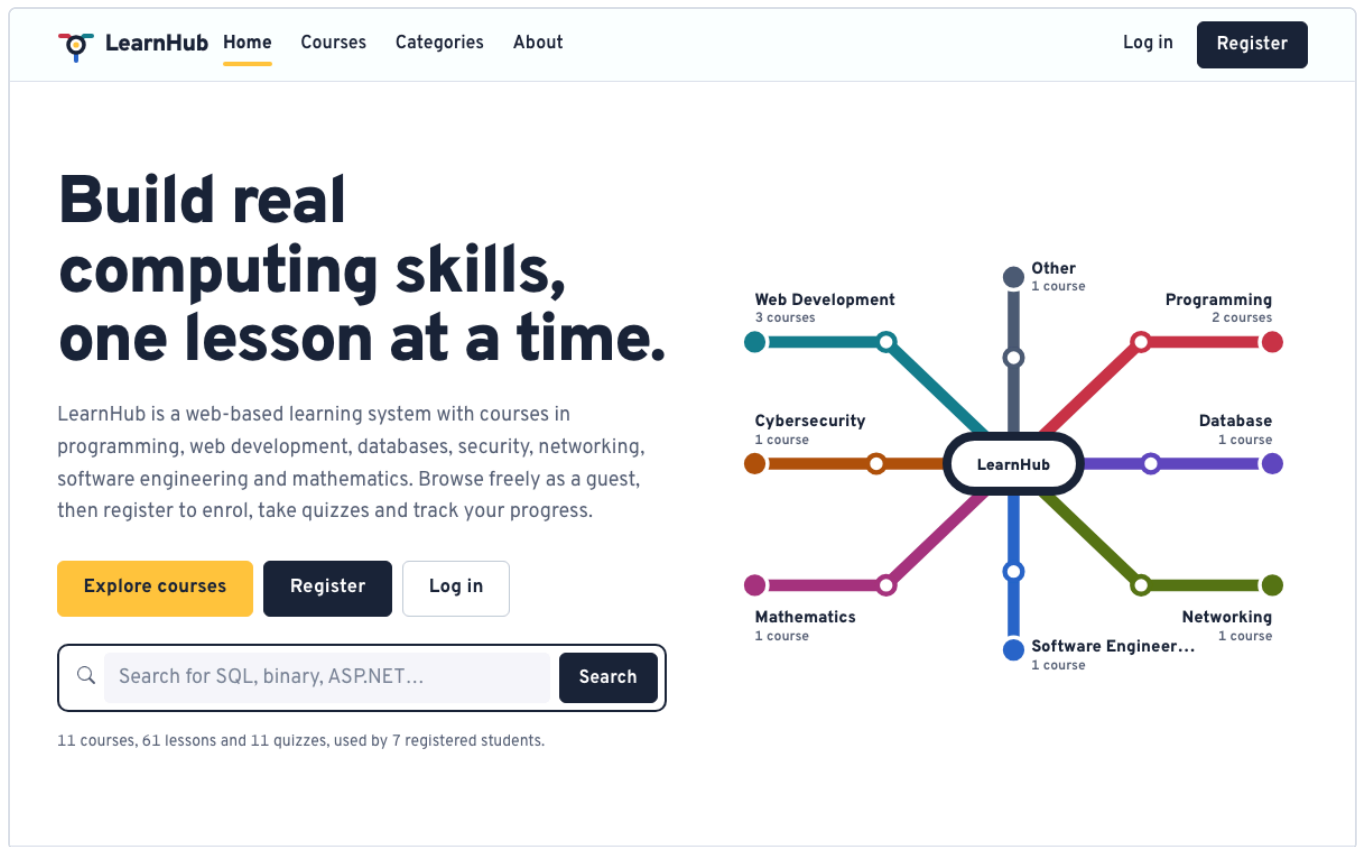
## 6. Deployment

- **Source and delivery:** the code lives on GitHub, which is both the source of truth and the CI runner. `ci.yml` runs three jobs on every push: build and the whole .NET test suite; the container job that builds the image and runs it the way the host will; and the Playwright browser suite against a production instance over HTTPS.
- **Application host:** Railway builds the multi-stage `Dockerfile` — a .NET 10 SDK stage that publishes the application and an ASP.NET Core 10 runtime stage that runs it — and `railway.json` selects the Dockerfile builder, checks `/health` with a 300-second timeout and restarts the service on failure. Railway supplies `PORT`, and the application reads it so that it listens on `0.0.0.0:<PORT>`; the image sets `PORT=8080` only as a fallback for local runs.
- **Database:** SQLite, with the file on a Railway volume mounted at `/data` and its path supplied as `DATABASE_CONNECTION_STRING=Data Source=/data/learnhub.db`. Migrations, roles, the administrator and the demonstration catalogue are applied at start-up, so the image needs no separate migration step.
- **Persistent state:** the same volume holds the Data Protection key ring under `DataProtection__KeysPath=/data/keys` and administrator uploads under `Storage__RootPath=/data/storage`. Without a volume every redeploy would start from an empty database, sign every user out and lose the uploaded files.
- **Proxy and TLS:** Railway terminates TLS at its edge proxy and forwards plain HTTP with `X-Forwarded-Proto` and `X-Forwarded-For`. `UsePlatformProxyHeaders()` turns those headers into `Request.IsHttps` and the real client address, because the framework does not read them on its own; without it HSTS would never be sent and secure cookies would be refused.
- **Presentation site:** <https://learnhub-wapp.web.app> is live on Firebase Hosting, served from `FirebaseLanding/` with `firebase.json` and `.firebaserc` at the repository root. Firebase cannot execute ASP.NET Core, so this is a static presentation and documentation page whose "Open Learning System" buttons link to the Railway application; after the first Railway deploy, `scripts/set-app-url.sh <railway-url>` rewrites that link and `firebase deploy --only hosting` republishes the site.
- **Status:** the Railway service has **not** been deployed yet — the CLI is installed but not authenticated. Every file and command is ready, and the container job in CI already proves the image, the migrations and the volume behaviour.

The step-by-step guide, configuration reference, verification checklist and rollback procedure are in [DEPLOYMENT.md](#).

## 7. Screenshots

### 7.1 Home page



**Build real computing skills, one lesson at a time.**

LearnHub is a web-based learning system with courses in programming, web development, databases, security, networking, software engineering and mathematics. Browse freely as a guest, then register to enrol, take quizzes and track your progress.

[Explore courses](#) [Register](#) [Log in](#)

Search for SQL, binary, ASP.NET... [Search](#)

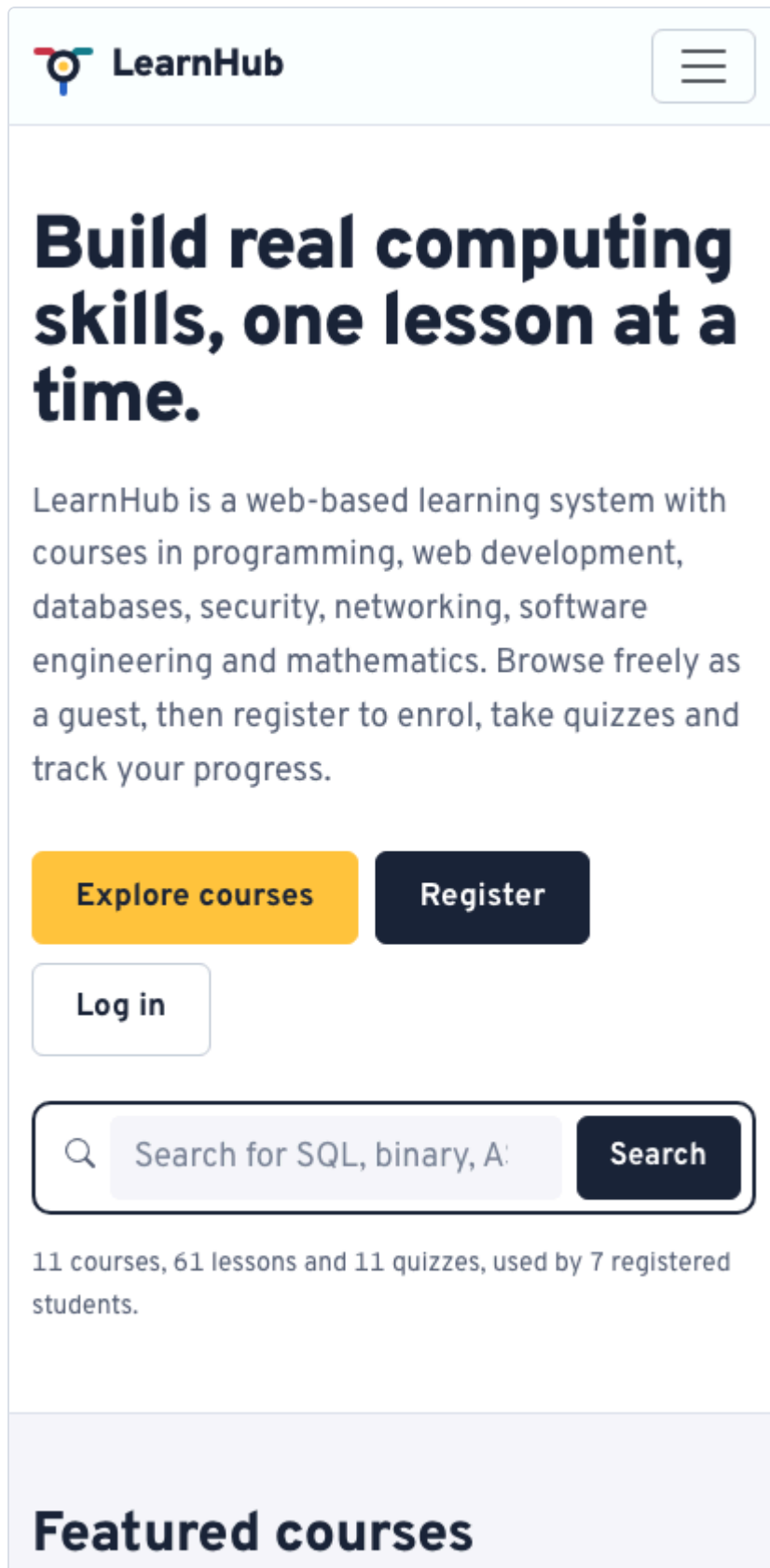
11 courses, 61 lessons and 11 quizzes, used by 7 registered students.

**LearnHub** categories:

- Web Development: 3 courses
- Cybersecurity: 1 course
- Mathematics: 1 course
- Other: 1 course
- Programming: 2 courses
- Database: 1 course
- Networking: 1 course
- Software Engineering: 1 course

The home page introduces LearnHub, offers a server-side course search and draws each category as a line into the LearnHub hub; every line links to that category's courses. Below are the most popular courses, how learning works and a call to action. Requirements: GUEST-01, UI-01, UI-02.

## 7.2 Home page on a phone



On small screens the navigation collapses behind a menu button and the map is replaced by category links, so the page stays fast and readable. Requirement: UI-05.

## 7.3 Course details

 **LearnHub** [Home](#) [Courses](#) [Categories](#) [About](#)

[Log in](#) [Register](#)

[Home](#) / [Courses](#) / [Web Development](#) / Modern HTML and CSS **Web Development**

# Modern HTML and CSS

Build accessible, responsive web pages with semantic HTML5 and modern CSS layout techniques.

● Beginner 🕒 9 h 📖 6 lessons 📝 1 quiz 👤 1 learner

👤 Taught by **Ms. Sofia Reyes**



### About this course

Every website starts with HTML for structure and CSS for presentation. This course teaches both the way professional front-end developers use them today.

You will write semantic HTML5 that works for screen readers and search engines, style pages with the box model, and build responsive layouts with flexbox and CSS grid. Along the way you will test pages on mobile, tablet and desktop screen sizes.

### Start learning

Create a free student account to enrol, track your progress and take quizzes.

[Create a free account](#)

[Log in to enrol](#)

Course details show difficulty, duration, lessons, quizzes, learners and instructor, the description and learning outcomes, and the course route. Guests can open the free preview lesson while the other stops are locked; the panel invites them to register or log in to enrol. Requirements: GUEST-06, GUEST-07.

## 7.4 Lesson player

The screenshot shows the LearnHub interface. At the top, there's a navigation bar with 'LearnHub' logo, 'Home', 'Courses' (highlighted), 'Categories', and 'About'. On the right, there are 'Log in' and 'Register' buttons. Below the navigation bar, a breadcrumb trail reads 'Courses / Modern HTML and CSS / How a web page is built'.

The main content area is divided into two columns. The left column, titled 'Course route', contains a vertical list of course topics: 'How a web page is built' (highlighted in yellow), 'HTML full tutorial for beginners', 'The CSS box model', 'CSS from zero to hero', 'Responsive layouts with flexbox and grid', and 'MDN: Learn web development'. A 'Course overview' button is at the bottom of this column. The right column displays the lesson content for 'How a web page is built'. It includes a 'Free preview' button, the lesson title, a brief description, a sub-header 'Three languages, three jobs', a bulleted list of topics (HTML, CSS, JavaScript), and a paragraph about maintaining separate files. Below this is a 'Semantic HTML5' section with a code example for HTML structure.

```
<header>
  <nav aria-label="Main">...</nav>
</header>
<main>
  <article>
```

An enrolled student reads a lesson with code examples beside the course outline, which shows completed and current stops. The course progress bar sits above the lesson, and the student can mark the lesson complete and move to the next one. Requirements: STU-08, STU-09, TECH-09.

## 7.5 Student dashboard

 **LearnHub** [Home](#) [Courses](#) [My Learning](#) [Quizzes](#) [Progress](#)

AR

[Profile](#) [Log out](#)

# Welcome back, Aisyah

You have completed 60% of your enrolled courses. Keep going.

[Browse courses](#)

3

Enrolled courses

1

Completed courses

12

Lessons completed

2

Quizzes passed

8

Courses you can still join

### Continue learning

Your next lesson in each course you are taking.

[All my courses](#)



**Relational Database Design with SQL**

12% complete

1 of 8 done

Next: SQL full database course

Resume



**ASP.NET Core MVC in Practice**

75% complete

6 of 8 done

Next: HTTP status codes reference

Resume

### Recent activity

✓

**Completed "Tables, keys and relationships"**

Jul 20, 2026, 08:14 AM

📅

**Enrolled in Relational Database Design with SQL**

Jul 20, 2026, 12:14 AM

🏆

**Scored 80% in MVC architecture quiz**

Jul 19, 2026, 02:14 AM

The dashboard shows figures from the database (enrolled and completed courses, lessons completed, quizzes passed, courses still available), courses in progress with Resume buttons, recent quiz results, recent activity and recommendations. Requirements: STU-01, STU-02, STU-03.

## 7.6 Quiz result


[Home](#)
[Courses](#)
[My Learning](#)
[Quizzes](#)
[Progress](#)

 Profile
 [Log out](#)

### Quiz results

Every quiz you have submitted, newest first.

2  
Attempts

2  
Passed attempts

90%  
Average score

| Quiz                    | Course                                       | Submitted              | Score                       |
|-------------------------|----------------------------------------------|------------------------|-----------------------------|
| MVC architecture quiz   | <a href="#">ASP.NET Core MVC in Practice</a> | Jul 19, 2026, 02:14 AM | 80% <a href="#">Review</a>  |
| HTML and CSS essentials | <a href="#">Modern HTML and CSS</a>          | Jul 13, 2026, 10:14 AM | 100% <a href="#">Review</a> |


 Structured courses, clear lessons and honest progress tracking for

**Explore**  
 All courses  
 Categories

**Account**  
 Profile  
 Privacy

**Project**  
 CT050-3-2-WAPP Web  
 Applications group  
 assignment Asia Pacific

After submitting, the student sees the score, whether the pass mark was reached, and every question with the chosen answer, the correct answer and an explanation. Requirements: STU-10, STU-11.

## 7.7 Admin dashboard


**LearnHub**
Admin

[View site](#)
 LearnHub Administrator

### Dashboard

Live figures for the LearnHub catalogue and learner activity.

[+ New quiz](#)
[+ New course](#)

|                                          |                                               |                                                  |
|------------------------------------------|-----------------------------------------------|--------------------------------------------------|
| <b>8</b><br>Users<br>7 students, 1 admin | <b>12</b><br>Courses<br>11 published, 1 draft | <b>8</b><br>Categories<br><a href="#">Manage</a> |
| <b>64</b><br>Resources<br>0 drafts       | <b>18</b><br>Enrolments<br>3 in 30 days       | <b>11</b><br>Quizzes<br><a href="#">Manage</a>   |

|                                                 |                                                      |                                                       |                                                           |
|-------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|
| <b>0</b><br>New students<br>in the last 30 days | <b>2</b><br>Course completions<br>enrolments at 100% | <b>12</b><br>Quiz attempts<br>67% passed, average 70% | <b>2</b><br>Unread messages<br><a href="#">Open inbox</a> |
|-------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|

**Most enrolled courses**

|                                     |            |
|-------------------------------------|------------|
| ASP.NET Core MVC in Practice        | 4 learners |
| Relational Database Design with SQL | 3 learners |
| JavaScript Essentials               | 2 learners |

**Enrolments by category**

|                 |              |
|-----------------|--------------|
| Web Development | 7 enrolments |
| Database        | 3 enrolments |
| Programming     | 3 enrolments |

[Quiz results](#)
[Messages](#)
[Main website](#)
[Log out](#)



The administrator sees live platform figures, the most enrolled courses, enrolments by category and recent activity in the protected admin area with its own navigation. Requirements: ADM-01, ADM-02.

## 7.8 Course management

**LearnHub Admin**

**Courses** [View site](#) **LA LearnHub Administrator**

**Courses**  
12 courses match the current filters. [+ New course](#)

Search:  Category:  Status:  [Apply](#) [Reset](#)

| Course                                                   | Level    | Status    | Lessons | Quizzes | Learners | Updated      |
|----------------------------------------------------------|----------|-----------|---------|---------|----------|--------------|
| <b>Effective Online Study Skills</b><br>Other            | Beginner | Published | 3       | 1       | 1        | Aug 27, 2026 |
| <b>Discrete Mathematics for Computing</b><br>Mathematics | Beginner | Published | 6       | 1       | 0        | Aug 18, 2026 |
| <b>Git and GitHub for Team Projects</b><br>Software      | Beginner | Draft     | 3       | 0       | 0        | Aug 9, 2026  |

The course list combines search and filters with cover thumbnails, level and publication status, counts of lessons, quizzes and learners, and edit and delete actions. Requirement: ADM-03.

## 7.9 Further screenshots to add

*Screenshot to add:* registration form showing validation messages (VAL-02, VAL-03). *Screenshot to add:* admin course form with cover upload (ADM-03, VAL-05). *Screenshot to add:* admin user details with role and deactivation actions (ADM-10). *Screenshot to add:* admin menu opened on a phone (UI-05).

Every GitHub Actions run also stores full-page screenshots of all browser scenarios on desktop, tablet and phone as the `e2e-screenshots` artifact.

## 8. Critical evaluation

### 8.1 Strengths

- **Complete journeys:** every role's journey works end to end, and automated tests prove it on the real database engine, inside the container the host will run and at three screen sizes.
- **Security by default:** access control, antiforgery, encoding, upload checks and secrets handling come from the framework and are tested with attack-style cases.
- **Truthful data model:** unique indexes, check constraints, deliberate delete rules, score snapshots and calculated progress.

- **Distinctive, accessible interface:** a consistent design system around one metaphor, with no automated accessibility violations.
- **Reproducible delivery:** one migration set, deterministic seed data, CI checks and a multi-stage `Dockerfile` mean anyone can rebuild the system from the repository and run the image the way the host will.

## 8.2 Limitations

- No email confirmation or password reset by email, and no multi-factor authentication.
- A single-instance design: migrations run at start-up, and the SQLite file, the key ring and the uploads share one mounted volume, so scaling out would need a different database and object storage.
- The Railway service is configured but not yet deployed, so the application has so far been exercised only locally, in the container job in CI and in the browser tests.
- Manual keyboard, screen-reader and real-device testing still needs to be completed by the team.

## 8.3 Lessons learned

- The assignment forbids Azure, so the earlier Azure deployment was replaced with Railway and Firebase. Reducing the data layer to a single SQLite provider at the same time removed a whole class of dialect differences from the test matrix and one entire schema to maintain.
- A migration that was never regenerated after a model change left the schema behind the model and broke 126 tests until it was recreated: a model change and its migration belong in the same change.
- Form pages failed on any host that terminates TLS in front of the application until forwarded headers were configured explicitly. `ASPNETCORE_FORWARDEDHEADERS_ENABLED` was an App Service hosting feature, and the framework does not read `X-Forwarded-Proto` on its own.
- Screenshots and browser tests at several sizes found layout bugs that unit and integration tests could not see.
- Designing from real content and one clear concept produced a more coherent interface than adding styles page by page.
- Framework security features, used consistently, are safer and simpler than custom code.

## 9. Future enhancements

1. Email confirmation, password reset and optional multi-factor authentication.
2. Certificates when a course is completed, and a public profile of achievements.
3. An instructor role that can manage only their own courses.
4. Question banks, randomised questions and timed quizzes.
5. Discussion threads per lesson, moderated by administrators.
6. A managed relational database and object storage for uploads, so the application can run more than one instance instead of sharing one mounted volume.
7. Monitoring and alerting for the deployed service, once the Railway deployment has been made.
8. Dependabot updates and vulnerability scanning in CI.

## 10. Conclusion

The team delivered LearnHub, a secure, responsive and tested learning management system that meets the assignment's requirements for guests, students and administrators. ASP.NET Core MVC, Entity Framework Core and Identity provided a solid, explainable foundation; a relational model with deliberate constraints keeps data correct; and a distinctive design system makes the product easy to use on any device.

Automated testing — 211 .NET test cases, a container check that proves the database survives a restart, and 42 browser and accessibility checks at three screen sizes — together with continuous integration gives confidence that the system works as described and can be maintained by future teams. The presentation site is live on Firebase Hosting, and the Railway service is built, configured and waiting for its first deploy.

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## Appendices

### Appendix A – Requirements traceability

Each requirement ID in [REQUIREMENTS\\_CHECKLIST.md](#) is mapped to its implementation, evidence and status in [REQUIREMENT\\_TRACEABILITY.md](#).

### Appendix B – Test run evidence

`.github/workflows/ci.yml` defines three jobs, all triggered by a push or a pull request:

| Job            | What it proves                                                                                                                                                                                                                                                                                                                                                            |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| build-and-test | The solution restores and builds with <code>-p:TreatWarningsAsErrors=true</code> , and all 211 .NET test cases pass on SQLite.                                                                                                                                                                                                                                            |
| container      | The Docker image builds, starts in Production with <code>PORT=8080</code> and a volume mounted at <code>/data</code> , answers <code>/health</code> , passes <code>scripts/smoke-test.sh</code> , logs that migrations were applied and the demo data seeded, and — after a restart — does <b>not</b> re-apply migrations, which proves the SQLite file really persisted. |
| e2e            | 14 Playwright scenarios × 3 viewports (42 checks) run against a production instance over HTTPS, including axe-core accessibility scans.                                                                                                                                                                                                                                   |

Each run stores the TRX result file, the container logs, the Playwright report, the full-page screenshots and the application log as downloadable artifacts. The figures in section 5 were measured on 14 September 2026 by running the same commands locally, and `.github/workflows/ef-migrations.yml` keeps the single SQLite migration regenerated on demand.

## Appendix C – User guide

**Guest:** open the site, search or pick a category on the Courses page, open a course and try a lesson marked "Free preview". Select **Register** to create an account. The illustrated walkthrough is in [USER\\_GUIDE.md](#).

**Student:** from the dashboard, select **Resume** on a course or open **Courses** and select **Enrol**. Work through the course route, select **Mark as complete** after each lesson, take the quiz when ready and review the explanations. Use **My Courses** for progress, **Quiz results** for history and **Profile** to change details or the password.

**Administrator:** log in to reach the admin dashboard. Use **Courses**, **Categories** and **Resources** to build the catalogue (drafts stay hidden until published), **Quizzes** to add questions, **Enrolments** and **Users** to manage people, **Quiz results** to review attempts and **Messages** for the contact inbox.

## Appendix D – Installation summary

With the .NET 10 SDK: clone the repository and run `dotnet run --project src/LearnHub`. The SQLite database is created, migrated and seeded automatically. In Development, strong random passwords for the DEMO ONLY administrator and student accounts are generated into the git-ignored `App_Data/demo-credentials.json`; set `Seed:AdminPassword` and `Seed:DemoStudentPassword` with `dotnet user-secrets` to choose your own. To run the image the way the host will, build the `Dockerfile` and mount a volume at `/data`; every variable is documented in `.env.example`. Full instructions are in the [README](#); production deployment is in [DEPLOYMENT.md](#).

## Appendix E – Individual contribution

| Member                             | Contribution summary                                                                                                           | Signature |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------|
| Khaytboy Khayrullaev<br>(TP072305) | Sole author: requirements, architecture, database, application code, tests, documentation and deployment — the complete system |           |